# T3 group factors: EFT_1_after_F

Active theory: SM + $S_1$ + $S_2$ + $C_{LLS_1S_2}$. The fermion $F$ is absent, so $y_1$, $y_2$ and $M_F$ are not running EFT couplings.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

# Notation key

| symbol | interaction / definition |
|---|---|
| $g_1, g_2, g_3$ | SM gauge interactions |
| $Y_u, Y_d, Y_e$ | SM Yukawa interactions |
| $y_1$ | $LFS_1$ + h.c. |
| $y_2$ | $LFS_2$ + h.c. |
| $M_F$ | $-\bar{F}M_F F$ |
| $T$ | $\mathrm{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$ | $\mathrm{Tr}(y_1^\dagger y_1)$ |
| $T_\nu^{(2)}$ | $\mathrm{Tr}(y_2^\dagger y_2)$ |
| $T_\nu$ | $T_\nu^{(1)} + T_\nu^{(2)}$ |
| $m_1^2, m_2^2$ | $m_1^2 S_1^\dagger S_1,\ m_2^2 S_2^\dagger S_2$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $\lambda_{S_i}^{(1)}$ | $\frac{1}{2}\lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$ |
| $\lambda_{HS_i}^{(1)}$ | $(H^\dagger H)(S_i^\dagger S_i)$ |
| $\lambda_{HS_i}^{(A)}$ | $(H^\dagger T^A H)(S_i^\dagger T^A S_i)$ |
| $\lambda_5$ | $HHS_1S_2^\dagger$ + h.c. |
| $\lambda_{12}^{(1)}$ | $(S_1^\dagger S_1)(S_2^\dagger S_2)$ |
| $\lambda_{12}^{(A)}$ | $(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$ |
| $\lambda_{12}^{(\times)}$ | $(S_1^\dagger S_2)(S_2^\dagger S_1)$ |

# Stage field content

| run | $S_1$ | $S_2$ | $F$ | integrated at this stage | active after matching |
|---|---|---|---|---|---|
| T3-C | $(2, \frac{1}{2})$ | $(2, \frac{3}{2})$ | $(3, 1)$ | F | S1, S2 |

### Field representations

| run | $S_1$ | $S_2$ | $\alpha$ |
|---|---|---|---|
| $\mathrm{T3-C}$ | $(2, \frac{1}{2})$ | $(2, \frac{3}{2})$ | 1 |

# Gauge and Yukawa beta functions in this EFT

$\beta_{g_1}^{(1)}$

| $\mathcal{S}$ | $\mathrm{T3-C}$ $(\alpha = 1)$ |
|---|---|
| $g_1^3$ | $\frac{17}{2}$ |

$\beta_{g_2}^{(1)}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $g_2^3$ | $-\frac{17}{6}$ |

$\beta_{g_3}^{(1)}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $g_3^3$ | $-7$ |

$\beta_{Y_u}^{(1)}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$ |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$ |
| $g_2^2(Y_u)_{ij}$ | $-\frac{9}{4}$ |
| $g_3^2(Y_u)_{ij}$ | $-8$ |
| $g_1^2(Y_u)_{ij}$ | $-\frac{17}{12}$ |
| $T(Y_u)_{ij}$ | $1$ |

$\beta_{Y_d}^{(1)}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$ |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$ |
| $g_2^2(Y_d)_{ij}$ | $-\frac{9}{4}$ |
| $g_3^2(Y_d)_{ij}$ | $-8$ |
| $g_1^2(Y_d)_{ij}$ | $-\frac{5}{12}$ |
| $T(Y_d)_{ij}$ | $1$ |

$\beta_{Y_e}^{(1)}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$ |
| $g_2^2(Y_e)_{ij}$ | $-\frac{9}{4}$ |
| $g_1^2(Y_e)_{ij}$ | $-\frac{15}{4}$ |
| $T(Y_e)_{ij}$ | $1$ |

$\beta_{m_1^2}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lambda_{S_1}^{(1)}\, m_1^2$ | $6$ |
| $g_2^2\, m_1^2$ | $-\frac{9}{2}$ |
| $g_1^2\, m_1^2$ | $-\frac{3}{2}$ |
| $\lambda_{12}^{(1)}\, m_2^2$ | $4$ |

$\beta_{m_2^2}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lambda_{S_2}^{(1)}\, m_2^2$ | $6$ |
| $g_2^2\, m_2^2$ | $-\frac{9}{2}$ |
| $g_1^2\, m_2^2$ | $-\frac{27}{2}$ |
| $\lambda_{12}^{(1)}\, m_1^2$ | $4$ |

$\beta_{\lambda_{HS_1}^{(1)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lvert\lambda_5\rvert^2$ | $6$ |
| $g_2^2\, \lambda_{HS_1}^{(1)}$ | $-9$ |
| $g_1^2\, \lambda_{HS_1}^{(1)}$ | $-3$ |
| $\lambda_1\, \lambda_{HS_1}^{(1)}$ | $6$ |
| $\left(\lambda_{HS_1}^{(1)}\right)^2$ | $4$ |
| $\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(1)}$ | $6$ |
| $\lambda_{12}^{(1)}\, \lambda_{HS_2}^{(1)}$ | $4$ |
| $g_2^4$ | $\frac{9}{4}$ |
| $g_1^4$ | $\frac{3}{4}$ |
| $\left(\lambda_{HS_1}^{(A)}\right)^2$ | $\frac{3}{4}$ |
| $\lambda_{HS_1}^{(1)}\, T$ | $2$ |

$\beta_{\lambda_{HS_2}^{(1)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lvert\lambda_5\rvert^2$ | $6$ |
| $g_2^2\, \lambda_{HS_2}^{(1)}$ | $-9$ |
| $g_1^2\, \lambda_{HS_2}^{(1)}$ | $-15$ |
| $\lambda_1\, \lambda_{HS_2}^{(1)}$ | $6$ |
| $\left(\lambda_{HS_2}^{(1)}\right)^2$ | $4$ |
| $\lambda_{HS_2}^{(1)}\, \lambda_{S_2}^{(1)}$ | $6$ |
| $\lambda_{12}^{(1)}\, \lambda_{HS_1}^{(1)}$ | $4$ |
| $g_2^4$ | $\frac{9}{4}$ |
| $g_1^4$ | $\frac{27}{4}$ |
| $\left(\lambda_{HS_2}^{(A)}\right)^2$ | $\frac{3}{4}$ |
| $\lambda_{HS_2}^{(1)}\, T$ | $2$ |

$\beta_{\lambda_{12}^{(1)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lvert\lambda_5\rvert^2$ | $3$ |
| $g_2^2\, \lambda_{12}^{(1)}$ | $-9$ |
| $g_1^2\, \lambda_{12}^{(1)}$ | $-15$ |
| $\left(\lambda_{12}^{(1)}\right)^2$ | $4$ |
| $\lambda_{12}^{(1)}\, \lambda_{S_1}^{(1)}$ | $6$ |

| | |
|---|---|
| $\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$ | 6 |
| $\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$ | 4 |
| $g_2^4$ | $\frac{9}{4}$ |
| $g_1^4$ | $\frac{27}{4}$ |
| $\left(\lambda_{12}^{(A)}\right)^2$ | $\frac{3}{4}$ |

$\beta_{\lambda_5}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lambda_5\,\lambda_1$ | 2 |
| $\lambda_5\,\lambda_{HS_1}^{(1)}$ | 4 |
| $\lambda_5\,\lambda_{HS_2}^{(1)}$ | 4 |
| $\lambda_5\,\lambda_{12}^{(1)}$ | 2 |
| $g_2^2\,\lambda_5$ | $-9$ |
| $g_1^2\,\lambda_5$ | $-9$ |
| $\lambda_5\,\lambda_{HS_1}^{(A)}$ | $-2$ |
| $\lambda_5\,\lambda_{HS_2}^{(A)}$ | 2 |
| $\lambda_5\,\lambda_{12}^{(A)}$ | $-\frac{1}{2}$ |
| $\lambda_5\,T$ | 2 |

## Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $g_1^2 g_2^2$ | 6 |

$\beta_{\lambda_{HS_2}^{(A)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $g_1^2 g_2^2$ | 18 |

$\beta_{\lambda_{12}^{(A)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $g_1^2 g_2^2$ | 18 |

## Full Weinberg-coefficient RGE in EFT1

Writing the canonical mixed LLSS coefficient as $C_{12}$, the coupled one-loop Weinberg equation used in the $F$-first EFT is

$$16\pi^2\frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \tag{1}$$

$$- \frac{3}{2}\left[(Y_e^\dagger Y_e)^T C_5 + C_5(Y_e^\dagger Y_e)\right] \tag{2}$$

$$+ R_W\,\lambda_5\,C_{12}. \tag{3}$$

Note: $\frac{7}{2}$ thing isnt there if we integrate out our fermion F so its not here

### Direct $C_{12} \to C_5$ recoupling factor

For each representation the source term is

$$16\pi^2\,\beta_{C_5} \supset R_W\,\lambda_5\,C_{12}.$$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $R_W$ | $-\frac{4}{3}$ |

.

$\mathrm{T3-C}\ (\alpha=1)$

*2 independent complex matched structures displayed.*

| complex matched operator | matched prefactor | $SU(2)_L$ invariant |
|---|---|---|
| $(L_i^T C L_j)\,S_1^\dagger\,S_2 + \mathrm{h.c.}$ | $-\frac{\sqrt{3}\overline{y_1}\,\overline{y_2}}{2M_F}$ | $\epsilon$ |
| $(L_i^T C L_j)\,S_1^\dagger\,S_2 + \mathrm{h.c.}$ | $-\frac{\overline{y_1}\,\overline{y_2}}{2M_F}$ | $\mathcal{I}_{LLS_1S_2}$ |

## Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched $C_5$ boundary condition and the subsequent SMEFT running is universal.